

Part 1 of 2

DESIGNING LOW-POWER PRODUCTS USING MSP430 MICROCONTROLLER



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The use of battery-operated electronics gadgets is increasing fast in our daily lives. Products are expected to be versatile, while consuming minimal power. Reduced power consumption has added advantages of smaller size and cost-effective products.

The MSP430 family of ultra-low-power microcontrollers from Texas Instruments meets these requirements well. These microcontrollers combine a number of precision components like comparators and analogue-to-digital converters with the versatility of a 16-bit CPU, Flash memory and many digital modules. The entire architecture is aimed at operating the end product at the lowest power without compromising on accuracy and functionality. Its low cost is an additional attribute that has seen the use of MSP430 microcontroller in products like blood-glucose meters, energy meters, clinical thermometers, handheld weather monitors and handheld precision instruments.

Here we introduce you to the architecture of MSP430 and tools to help you design ultra-low-power products.

Introduction to MSP430

Key features of MSP430 microcontroller are:

1. 16-bit RISC CPU
2. Current drawn is less than 0.1 μA in 'off' mode (RAM retention) and less than 0.5 μA in 'standby' mode
3. The microcontroller can wake up in less than 1 μs from power-down to full-speed functional mode
4. Zero-power-brownout detection circuit

5. 1kB to 128kB in-system-programmable (ISP) Flash memory

6. Intelligent peripherals to boost performance and versatility. These include:

- Timers
- ADC
- Comparators
- LCD drivers
- Communication ports like UART, SPI, I²C and IrDA

7. Embedded emulation

8. 14-pin to 100-pin packing options

The microcontroller features mul-

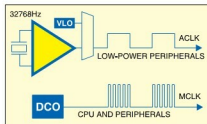


Fig. 1: Multiple-clock oscillator system

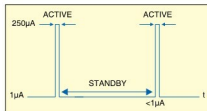


Fig. 2: Ultra-low-power activity profile

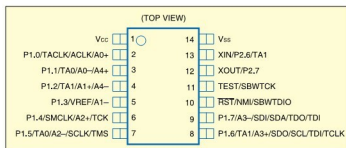


Fig. 3: Pin details of MSP430F213

tiple frequency clocks so as to run peripherals at the required frequency. There are three clocks in total, of which ACLK (auxiliary clock) is the low-frequency clock while MCLK (main clock) is the master clock which is normally at a higher frequency (see Fig. 1). The third is SMCLK (sub-main clock).

ACLK can be sourced from an external 32.768kHz watch crystal or an internal very low-frequency oscillator (VLO). The CPU is driven by MCLK, which is derived from the internal high-frequency digitally controlled oscillator (DCO) whose frequency can be selected using software settings.

MSP430 is designed to extensively support interrupt-driven burst processing as indicated in Figs 1 and 2. This ensures ultra-low-power operation of the system.

Key features of its 16-bit CPU are:

1. No accumulator bottleneck
2. RISC architecture with:
 - 27 core instructions
 - 24 emulated instructions
 - 7 addressing modes
 - Constant generator
3. Single-cycle register operations
4. Memory-to-memory atomic addressing
5. Bit, byte and word processing

In this part of the article, we have

described the Flash programmer-emulator for MSP430 family devices that can be built with readily available components. Part 2